

EXPERIMENT-12

16 BIT DIVISION

AIM: To write an assembly language program to implement 16 bit divided using 8086 processor.

Algorithm:

1. Read dividend (16) bit.
2. Read divisor.
3. Count $\leftarrow 8$.
4. Left shift dividend.
5. Subtract divisor from upper 8 bits of dividend.
6. If $CS = 1$ go to 9.
7. Restore dividend.
8. Increment lower 8 bits of dividend.
9. Count $\leftarrow$ Count - 1.
10. If count = 0 go to 5.
11. Store upper 8 bit dividend as remainder and lower 8 bit as quotient.
12. Stop.

PROGRAM:

MOV AX, [1100H] - load dividend

MOV BX, [1102H] - load divisor

DIV BX - Divide AX by BX

MOV [1200H], AX - store quotient

MOV [1202H], DX - store remainder

HLT - Halt program execution.

Input:

Address	Data
1100	10
1102	10

Output:

Address	Data
1200	1
1202	10

Result: Thus the program was executed successfully using 8086 processor simulator.

